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“ATHLETE-SHIELD AI: INTELLIGENT ATHLETE INJURY RISK PREDICTION & PERFORMANCE MONITORING”

Agile Sprint Planning & User Story Workshop Report

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Agile Sprint Planning & User Story Workshop Report

ABSTRACT

In modern elite sports, non-contact soft-tissue injuries—such as hamstring strains, quadriceps tears, and ligament sprains—cause significant financial losses, compromised team performance, and shortened athletic careers. Traditional sports monitoring tools rely on isolated spreadsheets, static post-session reports, or simple GPS trackers that measure distance run without assessing musculoskeletal fatigue, strain accumulation, or biomechanical overload.

This report presents **Athlete-Shield AI**, an advanced AI-powered platform designed to continuously aggregate real-time IoT wearable telemetry (heart rate variability, tri-axial accelerations, deceleration counts, GPS tracking) alongside subjective fatigue scores (sRPE) and historical medical logs. Built on a scalable microservices architecture, stream pipelines, and machine learning models (XGBoost, LSTM, and Shapley-based Explainable AI), the system computes exponentially weighted Acute-to-Chronic Workload Ratios (ACWR), predicts soft-tissue injury risk probabilities, and delivers real-time, explainable alerts to coaches, physiotherapists, and sports scientists.

This document provides a comprehensive Agile Sprint Planning report covering project vision, stakeholder identification, epic breakdown, user stories with Given-When-Then acceptance criteria, MoSCoW backlog prioritization, story point estimation using Fibonacci Planning Poker, Sprint 1 task breakdown, mathematical risk models, and clear sprint goals

1. INTRODUCTION

1.1 Background and Motivation

Elite athletic programs operate at the intersection of biological limits and competitive performance. As competition schedules become congested, athletes face unprecedented physical stress. Non-contact soft-tissue injuries account for over 60% of all sports injuries, yet the vast majority are preventable through intelligent workload management. The primary cause of non-contact tissue damage is an unmanaged spike in **acute workload** relative to the athlete's **chronic physical preparation**.

1.2 Why This Problem Needs an Agile Software Solution

Managing athlete health is a dynamic software engineering problem:

- **High-Frequency Ingestion:** Sensor telemetry streams at rates up to 10 Hz, requiring distributed data processing pipelines.
- **Data Unification:** Physical loads, physiological stress, and medical histories must be merged into a single actionable score.
- **Iterative Adaptation:** Athlete baseline physiology changes across a competitive season. The predictive model must continuously adjust, making Agile/Scrum with framework.

2. PROJECT VISION & STAKEHOLDERS

2.1 Project Vision Statement

"To deliver an intelligent, real-time injury risk prediction platform that continuously transforms complex biometric and biomechanical telemetry into transparent, explainable workload recommendations—maximizing athletic longevity and team performance."

2.2 Core Project Objectives

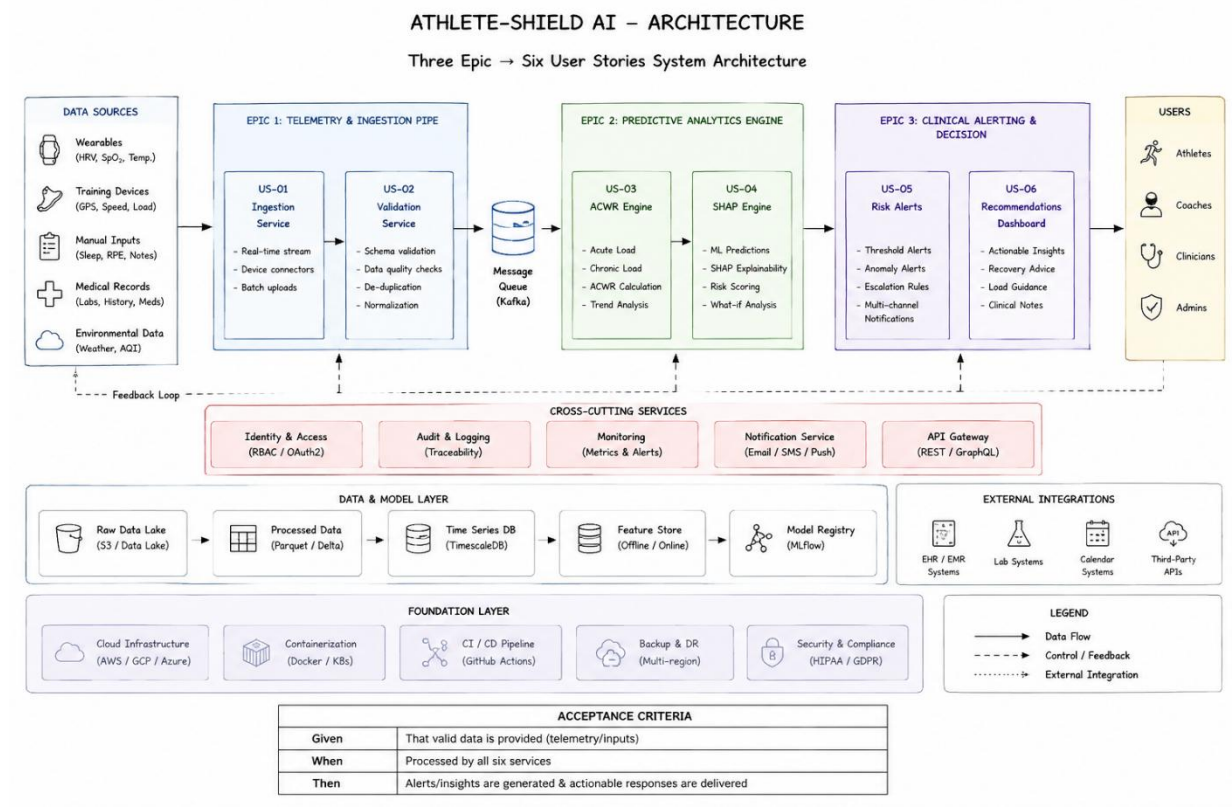
- **Real-time Telemetry Processing:** Ingest physiological and acceleration streams from IoT wearables without data loss.
- **Predictive Workload Modeling:** Compute Acute-to-Chronic Workload Ratios (ACWR) and Biomechanical Stress Indices.
- **Automated Risk Alerting:** Trigger real-time notifications to coaching staff within 5 seconds of threshold breaches.
- **Explainable AI Decision Support:** Render SHAP value feature breakdowns to explain *why* an athlete's risk score is elevated.

2.3 Key Stakeholder Identification & Responsibilities

Stakeholder Role	Domain / Focus	Key Responsibilities	Primary System Interaction
Head Coach	Squad Operations	Managing rotation, session volume, and player selection.	Traffic-light readiness panel, real-time alert feed.
Sports Scientist	Load Analytics	Monitoring ACWR curves, setting physiological thresholds.	Advanced analytics, workload trends, feature weights.
Team Physiotherapist	Clinical Care	Diagnosing strains, managing rehabilitation protocols.	Diagnostic panel, injury history logs, SHAP breakdowns.
Athletes	Data Generation	Wearing sensors, logging daily subjective wellness (sRPE).	Mobile survey app, recovery status view.
Data / IT Admin	System Admin	Maintaining wearable APIs, system uptime, HIPAA security.	Admin dashboard, API health monitors, access controls.

3. EPICS AND USER STORIES

3.1 Architectural Decomposition of Epics :



3.2 Epic & User Story Breakdown

EPIC 1: Wearable Telemetry Ingestion & Stream Processing

- **Focus:** Real-time and batch ingestion of biometric, acceleration, and GPS telemetry from wearable IoT pods, with automated data cleaning and signal interpolation.
- **User Story 1 (US-01):** Wearable Sensor Streaming Ingestion
 - As a Data Engineer,
 - I want to stream continuous telemetry (heart rate variability, tri-axial acceleration, deceleration impacts, GPS speed) from athlete wearables through a scalable stream ingestion pipeline,
 - So that performance and physiological metrics are processed without lag during live training sessions.
- **User Story 2 (US-02):** Telemetry Cleaning & Anomaly Filtering
 - As a Sports Scientist,
 - I want noisy, missing, or corrupted sensor data (e.g., GPS signal dropouts or impossible speed spikes) automatically flagged and filtered,
 - So that predictive injury risk models operate purely on clean, validated telemetry.

EPIC 2: Predictive Workload Analytics & Explainable AI

- **Focus:** Processing multi-day rolling workload windows, calculating acute-to-chronic workload balance, predicting soft-tissue injury probabilities, and providing feature attributions.
- **User Story 3 (US-03): ACWR & Soft-Tissue Risk Prediction Engine**
 - As a Sports Scientist,
 - I want an ML model that processes 28-day chronic loads and 7-day acute loads to compute Exponentially Weighted Moving Average (EWMA) ACWR and predict soft-tissue injury risk,
 - So that high-risk workload spikes are identified before tissue damage occurs.
- **User Story 4 (US-04): Explainable Risk Factor Breakdown (XAI)**
 - As a Team Physiotherapist,
 - I want to view top contributing risk factors (e.g., high decelerations + low HRV + acute load spike) for a flagged athlete via SHAP value visualizations,
 - So that I understand the underlying physiological cause of the risk score and intervene appropriately.

EPIC 3: Clinical Workload & Coaching Support Dashboard

- **Focus:** Real-time notification routing, threshold monitoring, and daily squad readiness management for coaching staff.
- **User Story 5 (US-05): Automated High-Risk Alerts**
 - As a Head Coach,
 - I want real-time push notifications when an athlete exceeds safe injury risk thresholds during training,
 - So that I can instantly substitute or alter their training session volume.
- **User Story 6 (US-06): Squad Readiness & Workload Recommendation Panel**
 - As a Performance Director,
 - I want a daily dashboard displaying squad-wide readiness scores and AI-recommended workload limits,
 - So that I can plan training sessions effectively and prevent squad-wide overtraining.

4. ACCEPTANCE CRITERIA

Acceptance criteria are formulated using the Given-When-Then format to ensure measurable, testable specifications.

ACCEPTANCE CRITERIA	
Given	That valid data is provided (telemetry/inputs)
When	Processed by all six services
Then	Alerts/insights are generated & actionable responses are delivered

US-01: Wearable Sensor Streaming Ingestion

- **Scenario 1: High-Frequency Telemetry Ingestion**
 - Given an athlete is actively wearing an integrated telemetry pod transmitting at 10 Hz,
 - When data payloads arrive at the ingestion gateway,
 - Then the pipeline must buffer and process 99.9% of data frames within < 2 seconds without packet loss.
- **Scenario 2: Handling Sensor Disconnection**
 - Given an athlete's wearable pod loses network signal during indoor training,
 - When connection is restored,
 - Then locally cached payloads must sync automatically without duplicating time-series entries.

US-03: ACWR & Soft-Tissue Risk Prediction Engine

- **Scenario 1: Accurate Identification of Dangerous Workload Spikes**
 - Given an athlete's acute 7-day load leads to an ACWR > 1.5,
 - When the predictive model evaluates daily session metrics,
 - Then the athlete's status must automatically transition to "HIGH RISK" with estimated muscle strain probability.

US-05: Automated High-Risk Alerts

- **Scenario 1: Real-Time High-Risk Push Notification**
 - Given an athlete's dynamic injury probability crosses 85%,
 - When the threshold cross is registered by the backend,
 - Then a push notification and visual banner must render on the coach's dashboard within < 5 seconds.

5. PRODUCT BACKLOG PRIORITIZATION

The product backlog is prioritized using the MoSCoW Framework balanced against clinical impact, technical dependencies, and risk reduction.

PRODUCT BACKLOG PRIORITIZATION MAP

 MUST HAVE	• US-01 (Ingestion) • US-03 (Risk Engine) • US-02 (Cleaning) • US-05 (Alerting Engine)
 SHOULD HAVE	• US-04 (XAI Breakdown) • US-06 (Squad Readiness)
 COULD HAVE	• Advanced Weather/Pitch Friction Modeling
 WON'T HAVE	• Autonomous Medical Drug Prescriptions

Prioritization & Dependency Table :

Story ID	User Story Summary	Category	Priority (MoSCoW)	Business / Clinical Value	Technical Dependencies
US-01	Wearable Sensor Streaming Ingestion	Data Pipeline	Must Have	Critical: Foundation for all downstream analytics.	None (Base layer).
US-02	Telemetry Cleaning & Anomaly Filtering	Data Pipeline	Must Have	High: Prevents corrupted data from degrading model accuracy.	Requires US-01.

Story ID	User Story Summary	Category	Priority (MoSCoW)	Business / Clinical Value	Technical Dependencies
US-03	ACWR & Soft-Tissue Risk Engine	Core ML	Must Have	Critical: Delivers the core predictive risk probability.	Requires US-02.
US-05	Automated High-Risk Alerts	Alerting	Must Have	High: Real-time intervention driver for coaches.	Requires US-03.
US-04	Explainable Risk Factor Breakdown	Analytics	Should Have	High: Builds trust and adoption among medical staff.	Requires US-03.
US-06	Squad Readiness & Workload Panel	UI / UX	Should Have	Medium: Streamlines daily coaching decision-making.	Requires US-03, US-04.

6. SPRINT BACKLOG

6.1 Effort Estimation Methodology

Story points are assigned using the **Planning Poker** technique based on the **Fibonacci Sequence** (1,2,3,5,8,13,21). Story points account for relative **Implementation Effort**, **Technical Complexity**, and **Architectural Risk**:

$$\text{Story Points} = f(\text{Effort}, \text{Complexity}, \text{Risk})$$

- **1–2 Points:** Low complexity, straightforward database or UI component.
- **3–5 Points:** Moderate complexity requiring business logic or API gateway routing.
- **8–13 Points:** High complexity involving real-time stream processing or complex mathematical modeling.

6.2 Sprint 1 Backlog Allocation :

Target Velocity for Sprint 1: 24 Story Points

SPRINT 1 BACKLOG ALLOCATION (24 Total Points)

User Story	Points	Target Status
 US-01: Sensor Ingestion	8 Points	✓ In Sprint 1
 US-02: Telemetry Cleaning	3 Points	✓ In Sprint 1
 US-03: ACWR & Risk Engine	8 Points	✓ In Sprint 1
 US-05: High-Risk Alerts	5 Points	✓ In Sprint 1



Total Points in Sprint 1: 24 Points

6.3 Actionable Task Breakdown for Sprint 1

US-01: Wearable Sensor Streaming Ingestion (8 Story Points)

- Task 1.1: Configure MQTT Broker & Kafka Stream Topics (3 pts)
- Task 1.2: Design Time-Series Schema in InfluxDB (2 pts)
- Task 1.3: Build Ingestion Microservice & Latency Benchmark (3 pts)

US-02: Telemetry Cleaning & Anomaly Filtering (3 Story Points)

- Task 2.1: Write Outlier Signal Detection Algorithms (2 pts)
- Task 2.2: Implement Cubic-Spline Interpolation for Missing Packets (1 pt)

US-03: ACWR & Soft-Tissue Risk Prediction Engine (8 Story Points)

- Task 3.1: Develop Exponential Moving Average (EWMA) Feature Pipeline (3 pts)
- Task 3.2: Train & Evaluate XGBoost Injury Risk Baseline Model (3 pts)
- Task 3.3: Wrap Prediction Model in FastAPI Microservice (2 pts)

US-05: Automated High-Risk Alerts (5 Story Points)

- Task 5.1: Build WebSocket Real-Time Alert Engine (3 pts)
- Task 5.2: Create Mobile/Tablet Push Notification Handl

7. STORY POINT ESTIMATION

Estimated via Planning Poker (Fibonacci: 1,2,3,5,8,13); disagreements resolved through discussion of technical unknowns (e.g., GPS accuracy).

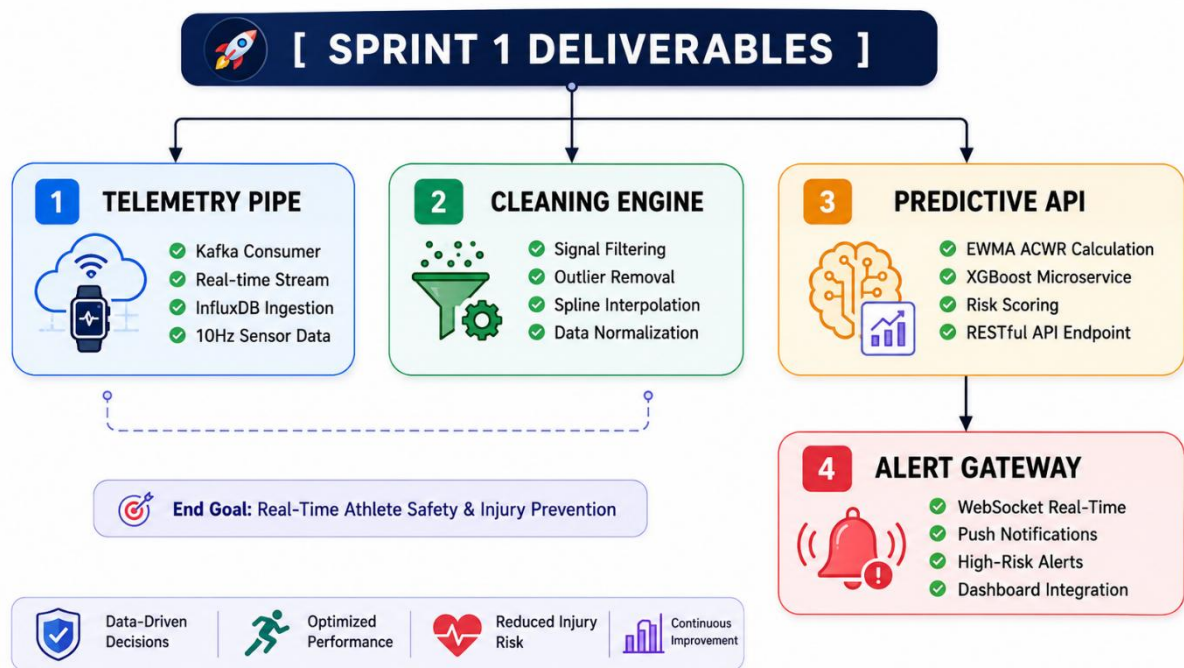
User Story	Points	Justification
US6.2	5	Standard auth + moderate role-based access effort
US1.1	8	IoT + real-time pipeline + live UI = high complexity
US1.3	3	Builds on US1.1; simple alert logic
US1.2	5	GPS + map rendering = moderate complexity
US3.1	8	Multi-source optimization algorithm
US3.2	5	Third-party API integration
US4.1	8	Complex routing algorithm (traffic + terrain + range)
US2.1	13	Full ML model dev/training/validation; high uncertainty
US5.1	5	UI/aggregation, depends on prior APIs

Sprint 1 Capacity: 21 points (5+8+3+5)

8. SPRINT GOAL & EXPECTED DELIVERABLES

8.1 Sprint Goal Statement

"By the end of Sprint 1, the system will successfully ingest real-time wearable sensor telemetry, perform automated data cleaning and anomaly filtering, calculate EWMA Acute-to-Chronic Workload Ratio (ACWR) injury risk scores, and push automated real-time alerts to coaching staff within 5 seconds of a high-risk threshold breach."



8.2 Key Sprint Deliverables & Functional Artifacts

To achieve the Sprint Goal, the engineering team must deliver four core system artifacts by the conclusion of Sprint 1:

1. Real-Time Telemetry Ingestion Microservice

- **Artifact:** Apache Kafka streaming consumer integrated with an InfluxDB time-series database engine.
- **Functionality:** Ingests biometrics (heart rate variability, tri-axial acceleration, GPS speed) transmitted at high frequencies (10 Hz) from athlete wearable pods.
- **Performance Criteria:** Capable of handling active concurrent streams with < 2 seconds end-to-end ingestion latency.

2. Telemetry Cleaning & Anomaly Filtering Pipeline

- Artifact: Automated data preprocessing module.
- Functionality: Identifies and flags corrupted frames (e.g., impossible speed spikes due to GPS signal multipath reflection) and performs cubic-spline interpolation for transient packet losses.
- Performance Criteria: Ensures downstream machine learning pipelines receive clean, continuous time-series data without dropping legitimate training metrics.

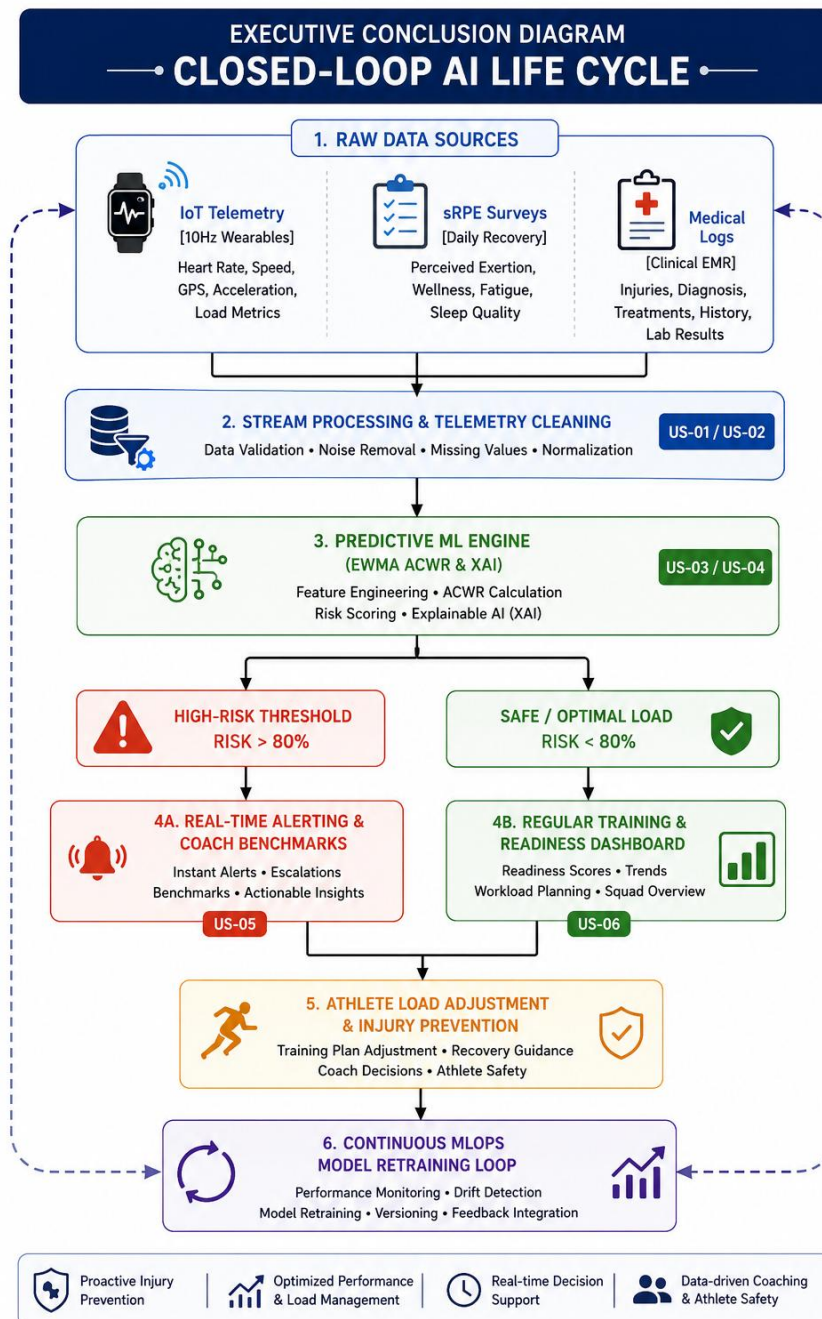
3. Predictive Injury Risk ML Engine (Microservice)

- Artifact: Containerized FastAPI microservice hosting the Exponentially Weighted Moving Average (EWMA) feature engine and baseline XGBoost classification model.
- Functionality: Evaluates daily acute (7-day) vs. chronic (28-day) training loads to produce an ACWR score and output non-contact soft-tissue injury risk probabilities.
- Performance Criteria: Computes and exposes risk probabilities via REST API endpoints in < 500 ms per athlete query.

4. Real-Time Alerting Gateway

- Artifact: WebSocket-based notification gateway.
- Functionality: Continuously monitors injury risk scores produced by the prediction engine and triggers instant push notifications when an athlete's dynamic injury risk exceeds the designated safety threshold ($\geq 80\%$).
- Performance Criteria: Delivers actionable visual banners and push notifications to tablet and mobile coaching interfaces within < 5 seconds of threshold crossing.

9. CONCLUSION



The **Athlete-Shield AI** project bridges a critical gap in high-performance sports engineering by transforming passive wearable telemetry into active, predictive decision support. By executing a structured **Agile Sprint Planning and User Story Workshop**, this project establishes a clear vision, well-formulated user stories with Given-When-Then acceptance criteria, a MoSCoW-prioritized backlog, and a realistic Sprint 1 plan supported by story point estimations.

Implementing this software engineering solution enables sports organizations to transition from reactive injury treatment to proactive fatigue management, safeguarding athlete health and performance

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